

Software Developer Practical Assessment

Task

Build working system that demonstrates practical senior-level engineering skills. The system should show clean backend structure, sensible data handling, basic AI/automation integration, and an interface or API that is easy to test.

Build an **Issue Intake and Smart Summary System** for a support or operations team.

- Users can submit an issue/ticket with fields such as title, description, priority, category, and status.
- The system stores the submitted records in a database.
- The system can list all submitted issues and filter them by status, category, or priority.
- Each issue should have a generated short summary and a suggested next action.
- The summary/next action may come from an LLM or from a well-structured rules-based fallback if no AI service is available.

Minimum Requirements

- Backend: Use the stack you are strongest in (e.g., Node.js, PHP, Python). Vanilla PHP or Laravel is preferred.
- Database: Use a relational or document database and explain why you chose it.
- API: Provide endpoints for create, list, view, and update issue records.
- Validation: Reject incomplete or invalid requests with proper error messages.
- Business logic: Add a simple rule such as overdue/high-priority issues being flagged for escalation.
- AI/Automation: Generate a short issue summary and suggested next step from the description.
- Frontend or usable test interface: A basic web UI is preferred, but a clean API demo is acceptable.

What to Submit

- Working source code
- README with setup steps
- Sample data or seed script
- Short explanation of architecture and key decisions
- Short note on what you would improve with more time

How the Test Will Be Evaluated

Area	What we will look for
Code quality	Structure, readability, naming, maintainability
Backend/API design	Clear routes, validation, error handling, update flow
Data handling	Reasonable schema/model design and proper storage
AI/automation layer	Useful summaries/actions, graceful fallback, practical integration
Problem solving	Good judgment, sensible assumptions, practical decisions

Communication

Clear README and explanation of trade-offs